Contact explains patterns of language change in Italo-Romance

Francesco Pinzin
Università di Padova
francesco.pinzin@unipd.it

Abstract: Language contact is a crucial factor determining language change. This study defines a novel methodology to quantify the extent of its relevance, testing it on the distribution of 23 syntactic features in the Italian dialectal environment. Two sets of data are compared, representing two diachronic stages separated by 70-90 years. This allows to estimate for each feature (i) the amount of change attested in this timespan, (ii) the amount of contact attested at the beginning of the period considered. A simple regression analysis shows that contact accounts for up to 66% ($R^2 = 0.66$; $r = 0.812$) of the diachronic variation, a highly significant result. Further post-hoc analyses indicate that some features change more/less than contact would predict. In this respect, the decisive factor is the structural compatibility among the languages in contact, defined as the degree of partial intra-linguistic overlap that feeds cross-linguistic influence in the attested contact situations (Hulk & Müller 2000). Given an equal amount of contact, features that are subject to more instances of intra-linguistic overlaps change more than others. Other factors, such as borrowability scales and areal asymmetries, do not play a significant role. This second result underlines the relevance for contact-induced change of individual-level processes active within multilingual speakers.

1. A test for the contact-change relationship

Contact between languages is a well-known mechanism triggering language change (Weinreich 1953, Thomason & Kaufman 1988, among many others). The extent of its relevance, however, remains more controversial, compared to other possible language-internal processes that operate during vertical transmission of the language across generations, such as contact-independent reanalysis and grammaticalisation (see Kuteva et al. 2020; see Roberts & Roussou 2003 for a formalisation in terms of "upward reanalysis", but see Weiß 2019, 2021 for cases of "downward reanalysis"). Also controversial is the extent to which contact is conditioned by language-internal grammatical factors. Thomason & Kaufmann (1988) review previous proposals on absolute and implicational grammar-internal constraints on the effects of contact, such as the need for a certain level of structural compatibility between the two languages (see Jakobson 1938; Weinreich 1953; see Aikhenvald 2006 for an overview). They conclude, however, that such constraints can be overcome, provided enough pressure. Beyond structural compatibility, another factor that is claimed to modulate contact is the inherent borrowability of different grammatical features: some features resist contact-induced change more than others. This is represented by so-called *borrowability scales*, i.e. scales defining which kinds of features are more prone to change under contact (see Gardani 2022 for a recent overview). For example, Thomason & Kaufmann (1988: 74-76) claim that content words are borrowed more easily than word-order rules, which in turn are borrowed more easily than word-structure rules. Such claims on structural compatibility and features' inherent borrowability still need to be extensively tested, however, especially within environments that allow controlling for other variables, such as sociolinguistic factors in the contact situation (e.g. language prestige and normativity), number of speakers, proficiency imbalances, etc. In this respect, contact across dialects varieties represent a privileged scenario, in which a high level of variation is attested within a generally uniform contact situation (see Section 4.5 for data in support; see Section 5 for a definition of the limits of the analysis of dialectal environments).

The aim of this contribution is to quantify to which extent contact explains patterns of diachronic change in the dialectal landscape of Italy. Furthermore, we also want to check whether some features change more than others and if structural compatibility and/or borrowability scales motivate such differences. The distribution of 23 syntactic features is analysed in three datasets (see Section 2.1 for details), which span over 70-100 years, from the beginning (t_0) to the end (t_1) of the 20th century. This allows us to estimate how much the distribution of each feature changes in this time-span. This measure is then correlated with a contact measure, defined on the geographical distribution of each feature at t_0 (both measures are detailed in Section 2.2). Our results show that change and contact are strongly correlated, to the extent that contact accounts for almost 66% of the diachronic signal ($R^2 = 0.656$; $r = 0.812$). However, it is also demonstrated that not all features behave equally. The main reason for this lies in the structural compatibility between the languages in contact, operationalised as the degree of partial intra-linguistic overlap (in the sense of Hulk & Müller 2000, Müller & Hulk 2001; see Section 4.3 for details) that feeds cross linguistic influence (CLI) effects in the attested contact situations. Given an equal amount of contact, features that are subject to more instances of intra-linguistic overlaps change more than others. Instead, more general borrowability scales do not play a role.

Our results show that contact is the main driver of change in the investigated linguistic environment. Additional modulation only comes from mechanisms known to be active at the individual level, such as the presence of partial overlap patterns supporting CLI effects. This backs the idea that contact-induced change at the societal level is a function of contact within the individual, i.e. multilingualism (already Weinreich 1953), and paves the way for further testing of the role that more general cognitive processes that are thought to underlie CLI effects have on language change. In this sense, a natural expansion is to assess the role of structural priming (see Jäger & Rosenbach 2008, Pickering & Garrod 2017 on the connection between priming and language change, as well as Kootstra & Sahin 2018, Jacob et al. 2025, Karkaletsou et al. 2025, Baroncini et al. 2026, and the collection in Engemann & Şafak 2026 for recent testing of the hypothesis).

The paper is organised as follows. Section 2 defines the methodology. Results are presented in Section 3. In Section 4, additional variables are considered and it is shown that the number of instances of partial overlap per feature is critical. Section 5 concludes.

2. Methodology

Since Séguy (1971), linguists have investigated dialects to probe the relationship between linguistic variation and geographical distance. The replicated finding is that there is a significant positive correlation between linguistic variation and logarithmic geographical distance (Nerbonne & Heeringa 2007, Nerbonne 2010, Nerbonne 2013, Jeszenszky et al. 2017 a.o.).[1] Depending on the dataset and the measures adopted,[2] the correlation ranges from $r \approx 0.4$ to $r \approx 0.7$ (but see Szmrecsanyi 2012). This crucially means that geographical distance accounts for 16% - 50% (R^2 0.16 - 0.5) of the overall linguistic variation found in the dataset. Since the measures adopted – synchronic linguistic variation and geographical proximity – can only approximate contact and change (and noise in the data is expected) these results

[1] Following the logarithmic curve, the positive effect of geographical distance on linguistic distance grows fast and then slows down. This is not going to be crucial in the following discussion.

[2] Linguistic distance is measured differently depending on the study. The studies reported in Nerbonne (2010, 2013) measure it as the number of substitutions needed to transform the phonetic transcription of word1 in Dialect1 into the phonetic transcription of word1 in Dialect2 (Levenshtein distance). Jeszenszky et al. (2017) define it instead based on 60 syntactic features. Geographical distance is usually measured in terms of Euclidean distance, but also travel times are adopted (Jeszenszky et al. 2017).

suggest that contact exerts a non-negligible influence on change, and that dialectal variation can be highly informative in this respect.[3] These approaches, however, aggregate features and cannot tell us if they are all equally susceptible to the change-contact relation. Hence, in this paper a different methodology is developed to quantify the relationship between contact and change, where individual features are the units of the analysis (see also Vergeiner et al. 2025 for a similar point regarding the limits of aggregating features). At its core, this methodology raises the following question: is there a positive correlation between how much a given feature F changes between two given points in time, call them t_0 and t_1, and how much F was subject to contact at t_0? This provides a measure of how much of the diachronic variation can be explained as an effect of contact (the R^2 of the simple linear regression). Since individual features are the units of the analysis, we can also check whether some features change significantly more or less than predicted.

The results of this methodology are informative for our research questions. A near-perfect correlation with no variation across features would mean that (in the present dataset) diachronic change is purely driven by horizontal contact across dialectal varieties. Moreover, this would tell us that contact has the same effect independently of the contact situation / kind of feature involved, in line with a strong version of the hypothesis in Thomason & Kaufmann (1988). A strong correlation ($r > 0,5$) would mean that contact is crucial but cannot by itself explain the whole attested diachronic variation. Further qualifications might come from modulating the effects of contact based on the features' inherent grammatical characteristics (as its position on the borrowability scales) and/or their contextual structural compatibility. Finally, finding a very weak or absent correlation would imply that in this dataset and with this methodology no significant conclusion regarding the contact-change relation can be drawn.

2.1 The datasets: corpora and features selected, tagging procedure

To test the relationship between contact and change via the proposed methodology, we need at least two dialectal datasets covering the same area at a certain distance in time. The Italian dialectal environment offers in this respect a perfect scenario: the AIS (*Sprach- und Sachatlas Italiens und der Südschweiz / Atlante italo-svizzero*; Jaberg & Jud 1928-1940; see also Loporcaro et al. 2021 for a digitised version, available at https://www.ais-reloaded.uzh.ch/) contains a set of data collected in the 1920s, while the ASIt (*Atlante Sintattico d'Italia*, available at http://asit.maldura.unipd.it/; see Benincà & Poletto 2007) and the Manzini & Savoia Corpus (MS henceforth; Savoia et al. 2026, available at https://zenodo.org/records/19352290 and based on Manzini & Savoia 2005a, 2005b, 2005c) contain two more recent sets of data. The ASIt data have been collected starting from the early 1990s (data collection still ongoing), while the MS data have been collected around the turn of the 21st century. Relevantly, the data collection methodology is similar across the three datasets, which warrants comparability. More specifically, all datasets contain dialectal sentences elicited via questionnaires, where the informants were either asked to translate a

[3] Geographical proximity does not necessarily mean language contact, nor contact is necessarily constrained by proximity. This is particularly true when means of communication / information can easily cross long distances (telephone, internet, radio, TV etc.). However, this issue is less prominent with dialects, which are generally only spoken in person and in the local environment (at least in the relevant time-frame). Note that also our methodology takes geographical proximity as a proxy for contact, inheriting these caveats.
It is also true that synchronic variation is not a perfect proxy for measuring the relationship between change and contact. Indeed, the synchronic distribution of a feature might result from a geographically contiguous spread that has been subsequently "broken" by the spread of other competitors. Hence, a discontinuous synchronic areal distribution does not necessarily signal that the given grammatical feature did not uniformly spread *via* contact within the geographic area. Our methodology departs from the use of synchronic variation and adopts a diachronic measure instead.

sentence, usually from Italian, or to name an object (only for the AIS data).[4] Since we will only focus on syntactic features, only translation data will be relevant.
The choice to investigate syntax is due to two factors. The first is the relative rarity of comparative dialectological studies adopting a quantitative perspective focusing on syntax (see Wieling & Nerbonne 2015:256-257; although the trend is increasing in the last years, see Vergeiner et al. 2025 for a recent study and an overview; see also van Craenenbroek & van Koppen 2025 for a review of quantitative studies focusing on grammatical representations). The second factor is practical: one of the available datasets, the ASIt, only focuses on (morpho)syntax and does not provide phonetic transcriptions.
Features' selection is based on availability; only features that were available in each dataset to a comparable degree were chosen. To give an example, the presence/absence of differential object marking, characteristic of many southern Italian varieties (see Rohlfs 1969: 7; Manzini & Savoia 2005b: 502-525), has not been included, since the AIS dataset does not provide adequate coverage for this feature. The list of investigated features is in Table 1.

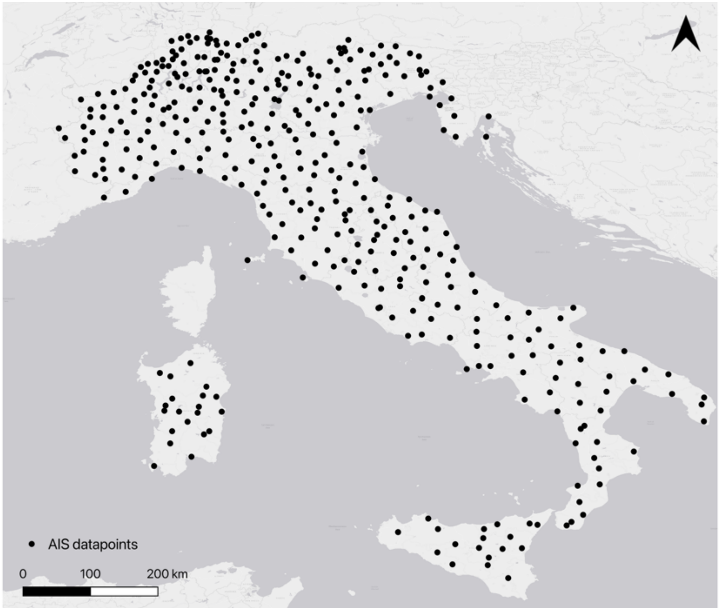
	Feature
1	Subject clitic for 2nd person
2	Subject clitic for 3rd person
3	Subject clitic for 3rd person, weather verb
4	Locative clitic
5	Partitive clitic
6	Postverbal wh items
7	Cleft structure with wh questions
8	Polar Q *che* 'that' marker
9	Doubly-filled Complementiser, main wh questions
10	Doubly-filled Complementiser, subordinate wh questions
11	Finite complementation with *fare*/*andare a* 'do/go to'
12	Finite complementation with modal verb *volere* 'want'
13	Noun-possessive order
14	Cooccurrence of definite article and possessive (global)
15	Cooccurrence of definite article and possessive (plural kinship nouns)
16	Cooccurrence of definite article and possessive (singular kinship nouns)
17	Indefinite objects realised as Partitive Articles
18	Auxiliary with reflexive *si* (*be* vs. *have*)
19	Preverbal sentential negation marker
20	Postverbal sentential negation marker
21	Use of *dove* 'where' for *da* 'from'
22	Synthetic past instead of analytic
23	Agreement between past participle and direct object (without object clitic)

Table 1. list of morphosyntactic features investigated

Considering the different characteristics of the original datasets, two partially different pipelines for tagging have been defined. The dialectal sentences in the AIS and the ASIt datasets can be grouped following the input sentences that have been provided to the informants as a basis for the translation. Hence, it was possible to define for each of these two datasets the set of informative sentences *per* feature F. The set is simply formed by all those sentences where the value for F can be defined. For preverbal and postverbal sentential

[4] While the AIS and the MS data have been collected via in person oral questionnaires, the ASIt data relies on a written translation questionnaire.

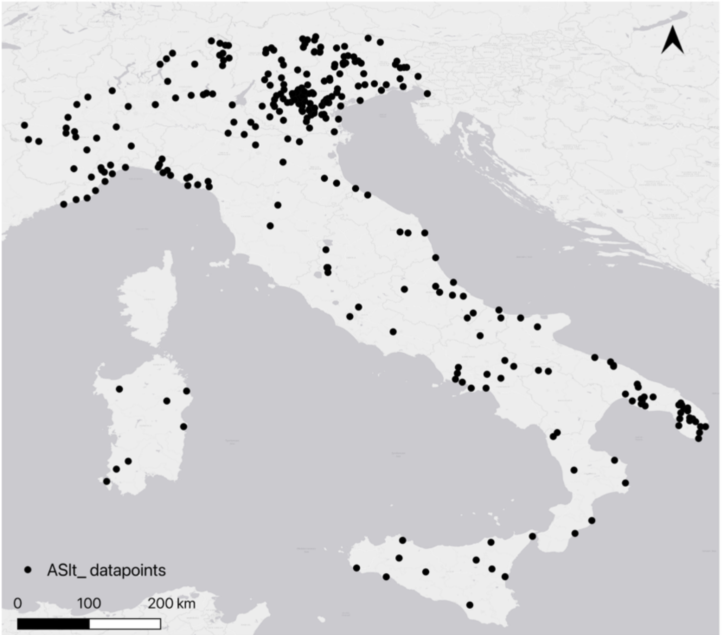
negation, for example, the set of input sentences containing sentential negation was selected.[5] Then, a spreadsheet was created for each F-dataset combination, resulting in 46 F-specific spreadsheets, 23 for the AIS, 23 for the ASIt.
In the AIS dataset, only one speaker is available for each of the 406 datapoints (i.e. geographical locations, plotted in Map 1). Thus, the AIS F-specific spreadsheets contain 406 rows, one for each datapoint, with columns marking source, name of the datapoint, geographical coordinates, administrative and dialectal categorisation (following Pellegrini 1977).



Map 1. AIS datapoints

In the ASIt dataset, instead, for some of the 289 datapoints (plotted in Map 2) more speakers are available. These are noted pairing the name of the variety and a number (e.g. for the variety of Borgonato, 7 speakers are available: from Borgonato1 to Borgonato7). The ASIt F-specific spreadsheets report then 428 rows, one for each speaker. The columns are the same as in the AIS spreadsheets, with an additional column reporting the speaker's code.

[5] The set of sentences investigated can be found in the following OSF repository https://osf.io/359rz/overview?view_only=5a49113cec064643b0fe03c64fd14c71, alongside all other relevant data and codes for replicating the experiment. All analyses have been defined and designed by the Author; AI tools (GitHub Copilot) have been used to check scripts and correct errors in them.



Map 2. ASIt datapoints

Each F-specific spreadsheet contains then columns for each sentence that is informative for F, where feature values are tagged. All features are binary (0,1), where 0 corresponds to the value that that feature has in Italian, 1 to the opposite value. To exemplify, let us take preverbal sentential negation. (1) reports three dialectal sentences translating the Italian input *non capisco* 'I don't understand', as well as their value.[6]

(1)

a. ***núŋ*** *gapísku* → 0
NEG understand.1SG
'I don't understand' (Acerno, Napoletano, ASIt)

b. *kapísį* ***mī́ga*** → 1
understand.1SG NEG
'I don't understand' (Albosaggia, Lombardo, ASIt)

c. *à* ***n*** *kapiž* ***brī́ža*** → 0
SBJ.CL NEG undesrstand.1SG NEG
'I don't understand' (Baura, Emiliano, ASIt)

Since Italian has a preverbal negative marker (*non* 'not'), 0 identifies all varieties where this is the case too, (1a) and (1c), while 1 identifies all varieties where the preverbal negative marker is not present, (1b). The tagging for postverbal sentential negation for the same set of sentences is reported in (2).

(2)

a. ***núŋ*** *gapísku* → 0
NEG understand.1SG
'I don't understand' (Acerno, Napoletano, ASIt)

b. *kapísį* ***mī́ga*** → 1
understand.1SG neg

[6] Dialectal sentences are reported following the original source. For details on the transcriptions adopted, refer to each source's documentation.

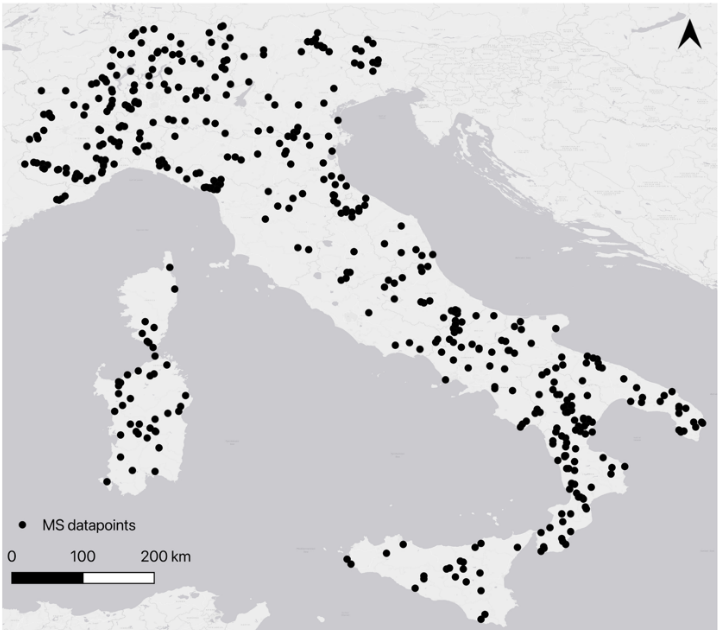
'I don't understand' (Albosaggia, Lombardo, ASIt)

c. *à* **n** *kapiž* ***bríža*** → 1
SBJ.CL NEG undesrstand.1SG NEG
'I don't understand' (Baura, Emiliano, ASIt)

Since Italian does not have a postverbal negative marker for standard sentential negation, 0 identifies all varieties where this is the case too (2a), while 1 identifies all varieties where a postverbal negative marker is present, (2b) and (2c).[7] The same applies to all features considered.[8]

Finally, each of the 46 F-specific spreadsheets reports a column with the average value for the relevant feature across all informative sentences. As detailed later, this average is the key value that will be imported for all computations.

The same methodology could not be replicated for the MS dataset, since it does not report the input sentences provided. As an alternative, sentences were grouped by datapoint. A single spreadsheet was created, with a row for each of the 489 datapoints, corresponding to 489 varieties (one informant per variety, as for the AIS), see Map 3.



Map 3. MS datapoints

The columns report the same general information added to the AIS and ASIt spreadsheets (name of the datapoint, geographical coordinates, administrative and dialectal categorisation). Then, a column for each of the 23 features was added. To tag the feature values for a given datapoint, the set of dialectal sentences for that datapoint was selected and then manually

[7] Note that preverbal and postverbal negation are independent features in the model. Varieties with discontinuous negation (as French *ne ... pas*) can be identified filtering for preverbal negation 0 & postverbal negation 1. This choice follows from the necessity to define binary features only.

[8] This classification raises problems concerning the use of cleft structures to form wh-Qs. The issue is that in standard Italian wh-Qs formed via cleft structures coexist with other structures, and regional variation is attested (cleft structures are more frequent in the north). We opted for treating clefts as non-standard, so that 1 identifies a cleft, while 0 identifies absence of a cleft. The same reasoning is extended to Partitive Articles with an indefinite interpretation (PAs). PAs are extensively used in Italian too, at variable degrees depending on the regional variety (see Cardinaletti & Giusti 2016, 2020 and ff.; Pinzin & Poletto 2021a, 2021b). Since their use is not equally widespread across the whole Italian area, we opted for marking absence of a PA as 0 ("Italian-like"). The issue is however immaterial to the present study, where change directionality is not investigated.

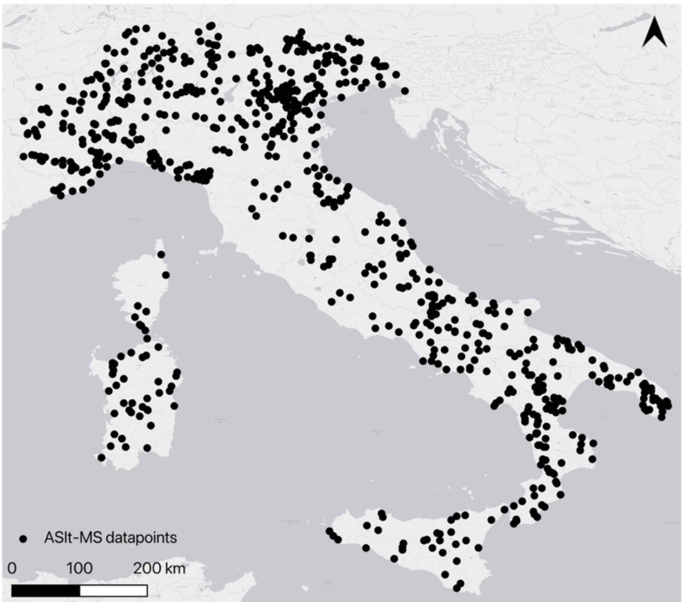
reviewed to detect informative sentences for each feature and, in case, which kind of value they showed. When contrasting values were found, an average across informative sentences was reported, matching the procedure adopted for the other datasets. While the tagging procedure for the AIS and ASIt data produced one spreadsheet per feature-dataset combination (46 spreadsheets), this tagging procedure produced one synthetic spreadsheet containing the average value for each feature across all datapoints.

2.2 Defining the measures for change and contact

The measure of change relies on detecting how much the distribution of each feature changed from t_0 (AIS) to t_1 (ASIt/MS).

The dataset for t_0 was derived merging in a synthetic spreadsheet the 23 F-specific spreadsheets defined based on the AIS data. This synthetic spreadsheet contains a column for each F and imports from the single-feature spreadsheets the average value of F across all informative sentences for each datapoint. This results in a single AIS dataset tracking the values of our 23 features across 406 datapoints, the t_0 reference.

To define the dataset for t_1, we merged the ASIt and the MS datasets in a single spreadsheet. To this end, we first merged in a single synthetic spreadsheet the 23 F-specific spreadsheets defined based on the ASIt data, following the same procedure adopted for the AIS data. Different speakers from the same datapoint were merged, averaging the value of F. This reduces the ASIt dataset to the same format of the other two: 1 row = 1 datapoint. Then, the MS dataset was merged with the ASIt dataset. Datapoints closer than 1km were merged, maintaining the ASIt ID and averaging across the two values found for each feature (29 merged datapoints).[9] The result is a single ASIt/MS dataset tracking the values of our 23 features across 749 datapoints, the t_1 reference (see Map 4).



Map 4. ASIt and MS datapoints

Next, we calculated how much each F changed between t_0 and t_1 by linking the AIS datapoints (t_0) to the ASIt/MS datapoints (t_1). Since the datapoints from the AIS and the ASIt/MS datasets do not fully overlap and some datapoints do overlap but have different names, a linking algorithm based on geographical proximity was defined. For each ASIt/MS

[9] Virtually no disagreement was found across the two datasets for these 29 datapoints.

datapoint, the linking algorithm searches the closest AIS datapoint, up to a maximum distance of 15km.[10] To avoid linking datapoints belonging to different linguistic groups, linking was restricted to datapoints with the same value for Area-Macro (defined following Pellegrini 1977). The result is a set of 531 pairs, covering 237 AIS datapoints. With this in place, the calculation of the mean diachronic change for a given feature F (Δ_F) is shown in (3).

(3)

$$\Delta_F = \frac{1}{|V_F|} \sum_{v \in V_F} |new_{F,d} - old_{F,d}|$$

For each linked datapoint d, subtract the value for F found in the ASIt/MS dataset ($old_{F,d}$) from the value for F found in the ASIt/MS dataset ($new_{F,d}$). Since we are not interested here in the directionality of change, the result of the subtraction is turned into an absolute value. The mean of the absolute result of the subtraction across all datapoints for which both $new_{F,d}$ and $old_{F,d}$ exist (V_F) is Δ_F, i.e. the estimation of how much F changed from t_0 to t_1.
Let us now clarify how feature values are imported for the computation. Initially, the average value of F across all informative sentences is imported (and, for the ASIt, across multiple speakers per datapoint). The level of informativity of this average value necessarily depends on the set of sentences chosen and their relative number. The datasets, however, do not allow for balancing these factors. Concretely, the ASIt requires to choose a large set of sentences to cover for the whole area, as different questionnaires have been used for different sub-areas. The same is not valid for the AIS, where usually only 1-2 input sentences *per* feature are available but they cover the whole area. To keep this potential imbalance in check, binarisations are adopted. Specifically, we set a series of 9 thresholds (from 0 to 0.8, increasing in steps of 0.1). Each threshold sets the lower boundary that turns the imported average to 1. For example, threshold 0 turns to 1 each imported average higher than 0, while threshold 0.8 turns to 1 each imported average higher than 0.8. Δ_F is independently computed for each binarisation and for the raw average value ($n10$ computations). The final value for Δ_F used for the analysis is the average across all computations; this minimises the effect of potential imbalances in the number and quality of the informative sentences tagged.[11]
Now that we have a measure for how much each feature changed from t_0 to t_1, the second step is to define how to calculate the extent to which each feature was subject to contact at t_0. This measure is only computed on the AIS dataset, so that no linking algorithm is in principle necessary. However, to maximise comparability with the change results, the computations for the contact measures are constrained to the 237 AIS datapoints that have a counterpart in the ASIt/MS dataset. We did that to avoid asymmetries in the areal coverage of the two measures: calculating the rate of contact on all AIS datapoints would track a part of contact that cannot be adequately tracked by our measure of change, which necessarily relies on the linking procedure.
Contact is operationalised as the number and intensity of contact opportunities between the two values of a given feature F. In other words, we count (i) how many datapoints with

[10] Different distances were tested. 15km was found to be the most restrictive distance that still allows to link a reasonable number of varieties (531 linked pairs, 237 AIS datapoints, almost 60% of the dataset). 10kms is too restrictive, as it only links 317 pairs and 186 AIS datapoints (around 46% of the dataset). Bigger distances risk linking varieties that already in the AIS dataset have different values for the investigated features. To further control for this risk, we also checked whether the amount of variation is significantly different for linked varieties that are closer than 1km and those that are further away. No significant difference was detected.
[11] A correlation analysis was performed (*cor*() function, *stats* package, R core team 2025; see OSF folder, fn. 5). The results show stability across computations (mean r = 0.888, only 1/45 pairs at 0.7).

different values for F are close to each other, (ii) how many close datapoints with different values for F are found on average per datapoint. The first contact measure is labelled 'breadth of contact' (B_F), the second 'intensity of contact' (I_F). For each F we calculate B_F and I_F as follows. We first set a list of radii, 25km, 50km and 75km.[12] The neighbourhood of a datapoint d relative to radius r ($N_r(d)$) is the set of datapoints d_1 belonging to the set of all datapoints (A) whose Euclidean distance e from d is smaller or equal to r and for which a value for F is attested, as reported in (4).

(4)
$$N_r(d) = \{d_1 \in A \mid d_1 \neq d, e(d, d_1) \leq r, F_{d_1} is\ attested\}$$

The set of contact neighbours of d for F and r ($C_{F,r}(d)$) comprises all datapoints d_1 in the neighbourhood set of d for which the absolute differential for the value of F is bigger than 0.1, see (5).[13]

(5)
$$C_{F,r}(d) = \{d_1 \in N_r(d) \mid |F_d - F_{d_1}| > 0.1\}$$

From the set of contact neighbours, breadth of contact for feature F and radius r ($B_{F,r}$) is the cardinality of the set of datapoints that both belong to the set of linked datapoints for F (L_F) and for which the set of contact neighbours is not empty (i.e. they have at least one neighbour that has a different value for F), see (6). Instead, intensity of contact for feature F and radius r ($I_{F,r}$) is the average across all datapoints in L_F of the number of neighbours for each datapoint, see (7).

(6)
$$B_{F,r} = |\{d \in L_F \mid C_{F,r}(d) \neq \emptyset\}|$$

(7)
$$I_{F,r} = \frac{1}{|L_F|} \sum_{d \in L_F} |C_{F,r}(d)|$$

Finally, B_F and I_F are averaged across the results obtained from each radius.
The results of the computation of the three measures (Δ_F, B_F and I_F) are in Table 2.

	feature	Δ_F	B_F	I_F
1	Sbj cl 2nd	0.0937	146.8333	1.0539
2	Sbj cl 3rd	0.0814	119.8667	0.8419
3	Sbj cl 3rd weather	0.222	154.0333	1.4322
4	Locative cl	0.1080	131	1.2217
5	Partitive cl	0.0990	111.9667	1.1681
6	Wh *in situ*	0.0456	0	0

[12] Larger radii would include datapoints for which contact is implausible to have taken place, while smaller radii would not include a meaningful number of datapoints. Crucially, the results of the contact measures for the three radii (crossed with the 10 threshold parameters) are highly correlated with each other, showing how the calculation is stable for this parameter (for I_F: mean r = 0.871, only 8/435 pairs below 0.7; for B_F: mean r = 0.855, 27/435 pairs below 0.7).

[13] The 0.1 limit is only relevant for the computation of the non-binary values, where the raw average is imported (1 out of 10 computations).

7	Wh cleft	0.0709	44.8667	0.3107
8	Polar Q *che*	0.0251	26.1	0.1540
9	Db-fill Comp main	0.1319	99.4	1.0819
10	Db-fill Comp sub	0.1943	175.4	2.3721
11	Finite compl *fare/andare a…*	0.0219	10.83333	0.0496
12	Finite compl *volere…*	0.011	5.7	0.0217
13	N-poss	0.0322	50.9	0.3496
14	Definite art & poss all Ns	0.0526	115.1333	1.1622
15	Definite art & poss pl kinship Ns	0.2089	137.8667	1.0333
16	Definite art & poss sg kinship Ns	0.1139	165.2667	2.079
17	Partitive Articles	0.1995	140.0333	1.3189
18	Aux *si*	0.2649	198.2	1.9829
19	PreV neg	0.0339	110.6333	1.2913
20	Post V neg	0.0738	110.9667	1.1338
21	*dove* for *da*	0.013	53.3667	0.3071
22	Synthetic past	0.0098	12.1	0.0698
23	Agr past part Obj	0.0803	52.7	0.3159

Table 2. Δ_F, B_F and I_F per feature

The next step is to check to which extent the contact measures can explain the change measure. To do that, we fit a simple linear regression. The analysis is performed in R (R version 4.5.2; R core team 2025) using the *lm*() function from the built-in *stats* package. Both predictors (B_F and I_F) are z-scored (mean = 0, SD = 1) prior to regression, so that their standardised coefficients (β) are directly comparable as effect sizes: each β expresses the change in Δ_F associated with a 1-SD increase in the predictor.

3. Results

The summary of the results from the two simple linear regressions between the contact measures (B_F and I_F, predictors) and the change measure (Δ_F, dependent variable) are in Table 3 and 4.

feature	Estimate (β)	Std. Error	t-value	Pr(>|t|)
(Intercept)	0.09511	0.01006	9.452	<.001 ***
I_F	0.05892	0.01029	5.726	<.001 ***

Table 3. Regression results, predictor I_F.

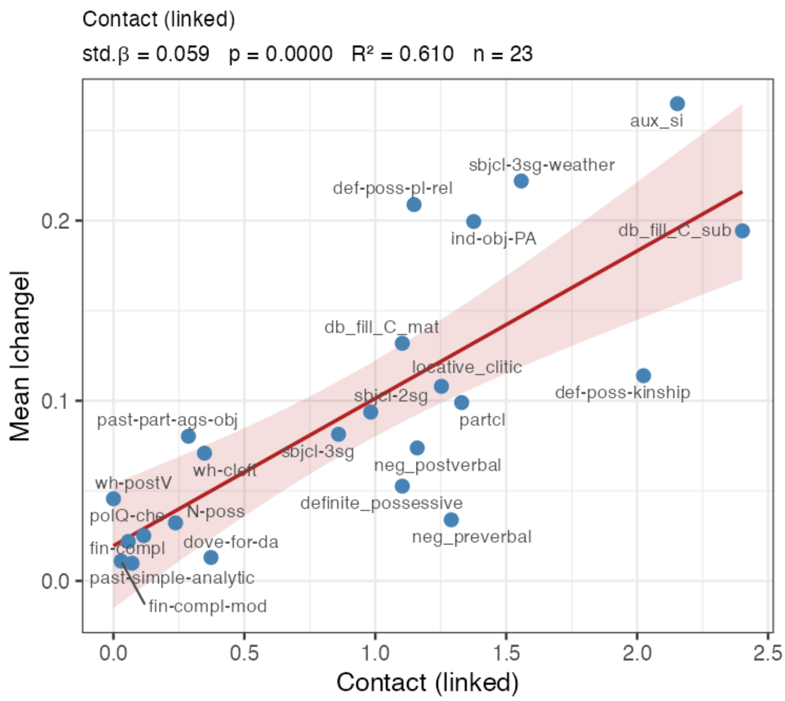
feature	Estimate (β)	Std. Error	t-value	Pr(>|t|)
(Intercept)	0.095113	0.009451	10.064	<.001 ***
B_F	0.061099	0.009663	6.323	<.001 ***

Table 4. Regression results, predictor B_F.

Both contact measures are highly significant positive predictors for the attested change: an increase in 1 SD in the predictor corresponds to an increase in Δ_F of about 0.059 for I_F and 0.061 for B_F. The more contact at t_0, the more change between t_0 and t_1.[14]

[14] The measure of contact for a given feature is constrained by the share of datapoints with the minority value. A feature with an extremely unbalanced distribution (e.g. 90-10) cannot have many datapoints where the opposite values are in contact. Hence, extremely unbalanced distributions are constrained to have low contact values (see also Kauhanen et al. 2021). Considering this, we investigated if F's initial skew in the distribution –

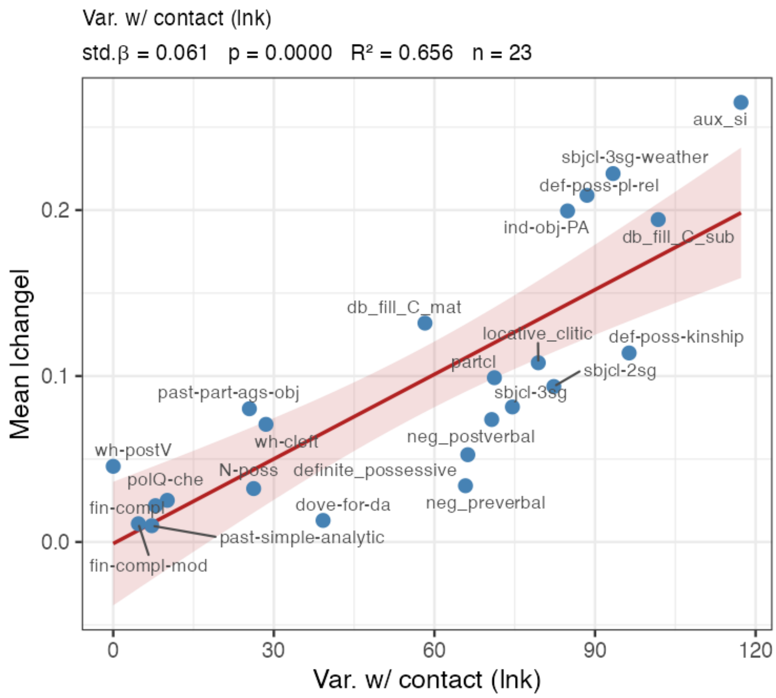
The best predictor is B_F. This is shown by the additional metrics of the two regressions. $I_F$explains 61% of the variance in Δ_F ($R^2 = 0.610$, adjusted $R^2 = 0.591$), while B_F explains 66% ($R^2 = 0.656$, adjusted $R^2 = 0.639$). The fact that the B_F is a better predictor than I_F is also visible in Plot 1 and 2, where the *x*-axis tracks the contact measure (I_F and B_F, respectively) and the *y*-axis the diachronic change measure (Δ_F). In Plot 2, the actual observations (one for each of the 23 features) lie closer to the regression line, minimising residual variation and improving fit.[15]



Plot 1. Regression results plotted for $\Delta_F \backsim I_F$.

implemented as the ratio of the minority value count to the majority value count, bounded at 1 – outperforms contact in predicting change. The results show that the initial skew, while significant, is not a real competitor of contact in predicting change (β = 0.036, p = 0.022, $R^2 = 0.226$). Moreover, this measure does not correlate significantly with the residuals of the contact-only model (Pearson r = -0.117, p = 0.5943). This shows that the driver of change is the spatial arrangement of opposite values, not the overall initial balance.

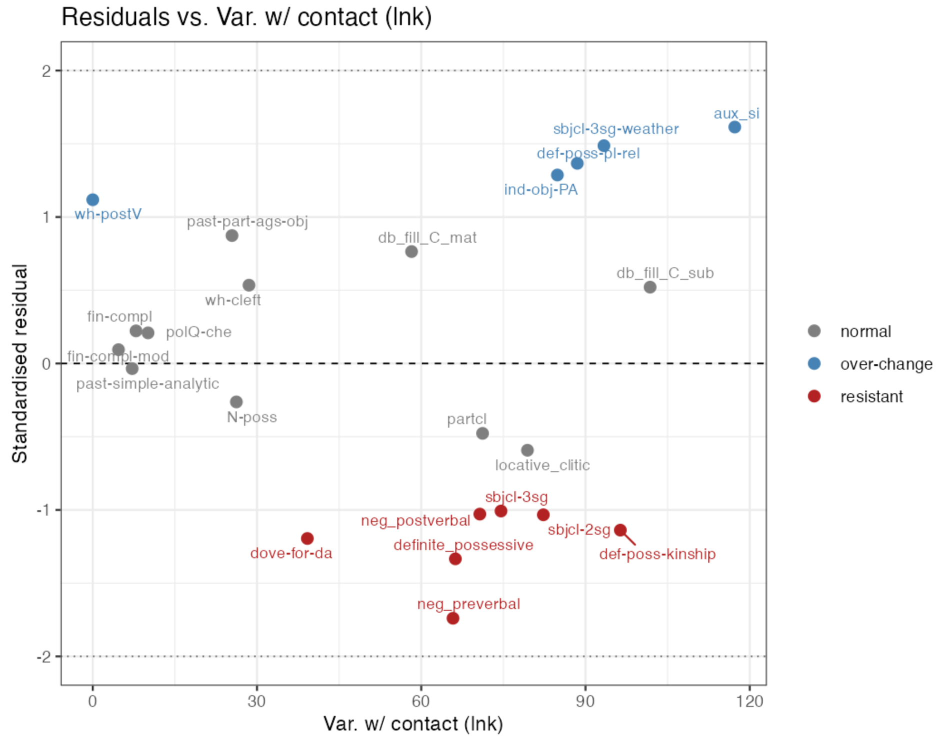
[15] While difference in fit between the two models is modest, two factors might explain why B_F tracks change better than I_F. Contrary to B_F, I_F is sensitive to the amount of contact per variety; its worse fit might then be connected to a ceiling effect: beyond a certain degree, an increase in the intensity of contact does not really lead to an increase in the chance of change. Moreover, B_F is also less sensitive to density imbalances in the areal coverage. Some areas present fewer datapoints per km^2 compared to other areas (see Map 4); varieties in more investigated areas therefore accumulate more counted neighbours, inflating I_F independently of a real increase in contact opportunity. B_F, a binary measure, is less affected by such imbalances.



Plot 2. Regression results plotted for $\Delta_F \backsim B_F$.

The visual analysis of the residuals (predictor B_F) shows a slight fan pattern (Plot 3), with residuals increasing with higher B_F values. However, the Cook's distance analysis, which reports how much the fitted values change when a given observation is deleted, highlights only auxiliary selection with reflexive *si* (*aux si*) as particularly influential. Removal of this feature does not however disrupt the results of the regression, nor its direction (explained variance decreased only marginally, $R^2 = 0.603$ from 0.656).[16]

[16] Results are reported in the OSF repository (fn. 5), along with a report on the sensitivity of the regression to the removal of this observation.



Plot 3. Plotted residuals for the regression $\Delta_F \backsim B_F$.

Plot 3 also highlights some features as "over-changers" (in blue; > 1 SD residuals), some others as "resistant" (in red, < -1 SD residuals).

- **Over-changers**: Wh *in situ*, Partitive Articles, definite + possessives with plural kinship Ns, 3rd persons subject clitics with weather verbs, use of be/have auxiliary with reflexive *si*.
- **Resistant**: *dove* for *da*, definite + possessives (general), postV negation, preV negation, 3rd person subject clitics, 2nd person subject clitics, definite + possessives with singular kinship Ns.

This indicates that we are in the second scenario defined in Section 2: contact is a fundamental driver of language change in the Italian dialectal environment. The more a feature is in contact at t_0, the more it changes between t_0 and t_1. The amount of change explained by our simple measure of contact significantly exceeds the average results in Nerbonne (2013) and related literature. In these studies, geographic signal is found to account for 16%-50% of the dataset's linguistic variation. The main reason for this difference is methodological: in the current implementation, the predicted variable is the attested diachronic change; in the mentioned studies, the predicted variable is the synchronic areal distribution of the phenomena. The former is a more direct measure of change than the latter, which captures the compounded effects of the set of subsequent changes that resulted in the current linguistic distribution of the phenomena.
However, while contact is crucial in defining how much a given feature changes, it does not explain the whole range of variation. We expand on this in the next section, where we investigate the potential role of additional factors.

4. Factors modulating contact-induced change: structural compatibility outweighs borrowability scales

4.2 Inherent borrowability
Following the literature, a factor to consider is the features' inherent level of resistance to contact-induced change. As already remarked in Section 1, the consensus is that some features change faster than others in contact situations. This observation is embodied by so-called borrowability scales. As a standard and well-known example, the scale from Thomason & Kaufman (1988) is reproduced in (8). Features higher in the scale can change also in low contact contexts, while features lower in the scale can only change in intense contact contexts. In other words, the lower a feature is on the scale the more it resists to contact-induced change.

(8)

Casual contact

Level 1	content words
Level 2	function words (conjunctions, adverbial particles)
	minor phonological features
	minor syntactic features
	minor lexical semantic features
Level 3	adpositions
	derivational suffixes
	phonemicization of previous allophones
	slight syntactic switches
Level 4	new phonemes
	word order changes
	inflectional borrowing
Level 5	new morphophonemic rules
	changes in word structure rules
	major changes in morphosyntax

Intense contact

(Thomason & Kaufmann 1988: 24; adapted from Gardani 2022: 867)

For the features analysed in this study, however, the role of such scales does not appear to be critical. This can be appreciated looking at the absence/presence of definite articles with possessives when used in combination with singular (*def-poss-kin*) or plural kinship Ns (*def-poss-pl-kin*), see (9) and (10).

(9)

a. *mẹ̀ kọžíne* (Albisano, Venetan, AIS)
my cousins

b. *lẹ mẹ̀ kọžínnę* (Crespadoro, Venetan, AIS)
the my cousins
'my cousins'

(10)

a. *tọ fr̥ḁ̊d̥ẹ̥́l* (Bagolino, Lombard, AIS)
your.SG brother

b. *ur tọ́ frędẹ́l* (Arcumeggia, Lombard, AIS)
the your.SG brother
'your brother'

These two features should belong to the same category, most probably Level 2 or 3 (minor syntactic features / slight syntactic switches). In both cases, the change involves the modification of a syntactic pattern within a lexically restricted category of Ns (kinship Ns, either singular or plural). Besides this, nothing changes in the distribution of possessives and definite articles. Nonetheless, (9) and (10) are on opposite sides: while changes from (9a) to (9b) occur more frequently than contact would predict, changes from (10a) to (10b) occur less often (see the position of *def-poss-kin* and *def-poss-pl-kin* in Plot 3). Now, consider *definite-possessive*. This feature tracks the presence/absence of definite determiners with regular Ns (11).

(11)

a. *vǫ̂sta vę̂sta* (Villafalletto, Piedmontese, AIS)
your.PL robe

b. *la vǫ̂sta vę̂stɑ* (Pianezza (Borgosesia), Piedmontese, AIS)
the your.PL robe
'your robe'

Changes between (11a) and (11b) imply a switch from "possessive determiners" – i.e. possessives that are inherently definite and cannot therefore cooccur with any determiner, as in French – to "possessive adjectives" – i.e. possessives that follow definite and indefinite determiners, as in Italian – or *vice versa* (see Pescarini 2025 for an in-depth overview). Intuitively, changes concerning (11) should be more demanding that changes concerning (9)-(10), as they concern the distribution of the whole class of possessive items. Nonetheless, our data show that (11) resists change as much as (10), and both are opposed to (9).
The same reasoning extends to subject clitics: while presence/absence of 2nd and 3rd singular subject clitics with referential subjects – see (12)-(13) – change less than contact would predict, 3rd singular subject clitics with weather verbs – see (14) – follow the opposite pattern.

(12)

a. *t ǫ̀y küžī́ bę́ṇe* (Calizzano, Ligurian, AIS)
SBJ.CL.2SG have.2SG sewed well

b. *áy kižī́u bę́ŋ* (Zoagli, Ligurian, AIS)
have.2SG sewed well
'you sewed well'

(13)

a. *ęl zę̀ vẹstī́ọ* (Cavarzere, Venetan, AIS)
SBJ.CL.3SG is.3SG dressed

b. *ę̀ vẹstī́ọ* (Teolo, Veneto, AIS)
is.3SG dressed
'he is dressed'

(14)

a. *l à pyụvǫ́* (Bologna, Emilian, AIS)
SBJ.CL.3SG has.3SG rained

b. *a pyụvúw / pyuvę́(ɑ)ṣt* (Comacchio, Emiliano, AIS)
has.3SG rained
'It rained'

Finally, consider sentential negation. This phenomenon too is split across preverbal and postverbal negation, see (1) and (2). While both features are categorised as "resistant" to

change, it is apparent from Plot 3 how preverbal negation resists change much more than postverbal negation.
Considering this evidence, inherent borrowability scales such as (8) does not appear to play a decisive role in discerning between the different propensity of our features to change under contact.

4.2 The overlap condition on cross-linguistic influence as a measure of structural compatibility

Let us now evaluate the role of structural compatibility. To operationalise this notion, we turn to the literature on cross-linguistic influence (CLI), an individual-level process active within multilingual speakers. A well-known condition on CLI is the degree of *partial surface overlap* between the two languages (see Hulk & Müller 2000, Müller & Hulk 2001 ff.; see Bidese & Tomaselli 2021, Bidese 2023, for an application of this concept to contact in the Italian linguistic environment). Let us see how this condition can be adopted as an operationalisation of structural compatibility, starting from its definition. Let *c* be a context where language A adopts structure *x*, while language B adopts structure *y*. For structure *x* to be transferred from A to B at the expenses of *y* in *c*, B must also exhibit a structure in its grammar that – at least on the surface – resembles *x* (albeit necessarily not used in *c*). This structure acts as a language-internal support for the transfer of *x* from A to B in *c*, at the expenses of *y*. As an example of an overlap situation, Müller & Hulk (2001) analyse object drop in bilingual children of Germanic (German/Dutch) and Romance (French/Italian) languages. While German and Dutch allow for topic-drop (15), French and Italian do not (16).

(15) *Habe* *ich* *gemacht.* (Ger.)
have.1SG I done
'I did that.'

(16) **J'* *ai* *fait.* (Fr.)
SBJ.CL.1SG have.1SG done
'I did that.'

However, both French and Italian exhibit structures where the canonical object position is left empty, either with object clitics (17) or with specific verbs like *savoir/sapere* 'to know' (18)-(19).

(17) *Je* *l'* *ai* *fait.* (Fr.)
SBJ.CL.1SG OBJ.CL have.1SG done
'I did that'

(18) *Je* *sais* (Fr.)
SBJ.CL.1SG know.1SG
'I know (that)'

(19) *Non* *so* (It.)
not know.1SG
'I don't know (that)'

Following the above definition, the surface absence of overt objects in (17)-(19) defines a situation of partial overlap, which would support CLI effects leading to the production of sentences like (16) in bilingual speakers of German/Dutch and French/Italian. Indeed, they find that the group of Germanic/Romance bilingual children produce sentences like (16) to a significantly higher rate than the group of Romance monolingual children. Instead, CLI effects are not found with root infinitives, where both Romance ad Germanic languages work similarly. Such a situation of full overlap does not clearly lead to any transfer between the two languages (on the definition of partial and full overlap, see also Unsworth 2003). Hulk & Müller (2000) and Müller & Hulk (2001) propose a strong version of the hypothesis, whereby CLI is only possible if the partial overlap condition is met.[17] However, further studies found evidence for CLI effects also without any overlap between the two languages (already Yip & Matthews 2000, among many others; see Serratrice 2013 for an overview). The hypothesis must then be weakened: partial overlap is not a necessary but a *facilitating* condition for CLI. Considering this, we can operationalise structural compatibility as the degree of partial overlaps for our features and check if it can explain the residual differential in their attested change.

Let us first reconsider presence/absence of definite articles with possessives, which is tracked by three features, see (9)–(11). Since each feature is binary, we have in principle 8 (= 2^3) plausible varieties. However, the space of variation is constrained by an implicational scale. Varieties adopting the DEF(INITE)+POSS(ESSIVE) pattern with plural kinship Ns (9b) are a subset of those adopting it with singular kinship Ns (10b). Furthermore, if a variety uses possessives without determiners – the POSS pattern – with regular Ns (11a) (i.e. if a variety has "possessive determiners", see Section 4.2), the same will also be valid for both singular and plural kinship Ns. Hence, there are only four types of varieties: L_A where DEF+POSS extends to all contexts (20), L_B where DEF+POSS extends to plural kinship Ns but not to singular kinship Ns (21), L_C where DEF+POSS is confined to regular Ns (22), and L_D where only POSS is attested (23).

(20)

a. *ul vǫ́z vęštī́d* (Arcumeggia, Lombard, AIS) L_A
the your.PL dress
'your dress'

b. *i mę́ küzíṇ*
the my cousins
'my cousins'

c. *ę mí küzī́nα*
the my cousin
'my cousin'

(21)

a. *αl vǫ́z vistí* (Cozzo, Lombard, AIS) L_B
the your.PL dress
'your dress'

b. *i mę́ küẓíɳi*
the my cousins
'my cousins'

c. *mę́ küẓíɳα*
my cousin
'my cousin'

(22)

a. *ęl vọ́ṣtrọ vẹṣtī́tọ* (Cerea, Venetan, AIS) L_C
the your.PL dress
'your dress'

[17] In addition to a so-called "interface condition", which dictates that CLI can only occur at the interface between two modules of the grammar, and more specifically at the interface between syntax and pragmatics (on this, see also Sorace & Filiaci 2006).

b. *mę̀ kuẓínẹ*
my cousins
'my cousins'
c. *mè kuẓína*
my cousin
'my cousin'

(23)
a. *vọ́sɑ mɑndū́rɑ* (Vrin, Rhaeto-Romance, AIS) L_D
your.PL dress
'my dress'
b. *máyɑṣ kuzɑrī́nɑṣ*
my cousins
'my cousins'
c. *máyɑ kuzɑrī́nɑ*
my cousin
'my cousin'

The contact situations among L_A, L_B, L_C and L_D are defined in (24)-(29) (where arrow = CLI supported by an overlap; bold = overlap). In the L_A-L_B contact situation (24), transfer *via* CLI of DEF+POSS from L_A to L_B in the context of singular kinship Ns is supported by an overlap. This is because L_B already has DEF+POSS in other two cells. The L_A-L_C contact situation (25) presents two patterns of CLI supported by an overlap, both involving transfer of DEF+POSS from L_A to L_C. Note that given the implicational scale, transfer should affect more plural than singular kinship Ns (hence, the dotted arrow). The L_B-L_C contact situation (26) shows two patterns of CLI supported by an overlap, both involving plural kinship Ns: not only the extension of DEF+POSS from L_B to L_C is supported by the existence in L_C of DEF+POSS in another cell, but also the extension of POSS from L_C to L_B is supported by the existence in L_B of POSS in another cell. L_B-L_C contact should then give rise to a high degree of variability in the contexts of plural kinship Ns. L_D-L_A contact (27) shows not supported CLI pattern, while (28) and (29) mirror the patterns already described for (25) and (24), respectively.

(24)

	L_A (Arcumeggia type)		L_B (Cozzo type)
Regular Ns	DEF+POSS		**DEF+POSS**
Pl kinship	DEF+POSS		**DEF+POSS**
Sg kinship	DEF+POSS	→	POSS

(25)

	L_A (Arcumeggia type)		L_C (Cerea type)
Regular Ns	DEF+POSS		**DEF+POSS**
Pl kinship	DEF+POSS	→	POSS
Sg kinship	DEF+POSS	⋯→	POSS

(26)

	L_B (Cozzo type)		L_C (Cerea type)
Regular Ns	DEF+POSS		**DEF+POSS**
Pl kinship	DEF+POSS	⇄	POSS
Sg kinship	**POSS**		POSS

(27)

	L_D (Vrin type)	L_A (Arcumeggia type)
Regular Ns	POSS	DEF+POSS
Pl kinship	POSS	DEF+POSS
Sg kinship	POSS	DEF+POSS

(28)

	L_D (Vrin type)		L_B (Cozzo type)
Regular Ns	POSS	⇢	DEF+POSS
Pl kinship	POSS	→	DEF+POSS
Sg kinship	POSS		**POSS**

(29)

	L_D (Vrin type)		L_C (Cerea type)
Regular Ns	POSS	→	DEF+POSS
Pl kinship	POSS		**POSS**
Sg kinship	POSS		**POSS**

These schemes capture an asymmetry between the three contexts, due to their relative position in the implicational scale. While both singular kinship Ns and regular Ns are subject to one full and one weakened overlap, plural kinship Ns are subject to four overlaps. Consequently, CLI in the context of plural kinship Ns should be more favoured than in the context of singular kinship Ns, leading to a higher degree of contact-induced change. Indeed, this is what we find in our data: plural kinship Ns change comparatively more than both singular kinship Ns and regular Ns, which instead behave similarly.

Let us now consider subject clitics. In this case too, an asymmetry emerges from the data: 3^{rd} and 2^{nd} singular referential subject clitics change less than contact would predict, while 3^{rd} singular subject clitics with weather verbs change more. Let us first focus on 3^{rd} singular subject clitics only. Here too, an implicational scale is present: varieties using 3^{rd} singular subject clitic with weather verbs are a subset of those using 3^{rd} singular subject clitic with referential subjects (see Poletto 2000 for an overview; see also Bidese & Tomaselli 2021, Tomaselli & Bidese 2023, Madaro et al. 2025 for recent assessments of contact in relation to subject clitics in the North-Eastern area of Italy). Hence, we have three types of languages: L_A, with a subject clitic in both contexts (30); L_B, with a subject clitic only with referential subjects (31); L_C, without a subject clitic in either of these contexts (32). (33)-(35) define the three contact situations and the partial overlaps.

(30)

a. *al* *à* *pyọvú* (Claut, Friulano, AIS) L_A
SBJ.CL.3SG has rained
'It rained'

b. *al* *ẹ̀* *vẹṣtí*
SBJ.CL.3SG is dressed
He is dressed'

(31)

a. *pyọvẹ̀ṣt* (Albisano (Torri del Benaco), Veneto, AIS) L_B
rained
'It rained'

b. *l* *ẹ̀* *vẹṣtí*
SBJ.CL.3SG is dressed
'He is dressed'

(32)

c. ę̀ *ppyọvū́to* (Gavorrano, Toscano, AIS) L_C
is rained
'It rained'

d. ę̀ *vvesṭī́to*
is dressed
'He is dressed'

(33)

	L_A (Claut type)		L_B (Albisano type)
ref	SBJ-CL+V		**SBJ-CL+V**
weather	SBJ-CL+V	⟶	V

(34)

	L_B (Albisano type)		L_C (Gavorrano type)
ref	SBJ-CL+V	⟵	V
weather	**V**		V

(35)

	L_A (Claut type)	L_C (Gavorrano type)
ref	SBJ-CL+V	V
weather	SBJ-CL+V	V

As *per* (33) and (34), both transfer from L_A to L_B of 3rd singular clitics in the context of weather verbs and transfer from L_C to L_B of the lack of 3rd singular clitics in referential contexts are supported by an intra-linguistic overlap. This conflicts with our findings, however, whereby we see 3rd singular clitics change much more in the context of weather verbs than in referential contexts. The issue is clarified once we check the areal distribution of L_A, L_B, and L_C (Map 4).[18]

[18] The AIS data report overt pronouns for 3rd person referential subjects in Sardinian varieties. This is most probably due how the data have been collected and reported, as Sardinian is a null-subject language (see Mensching & Remberger 2016 and references therein).

Map 4

Both L_A-L_B and L_A-L_C contact are frequently attested, while L_B-L_C contact is sporadic (if present at all). This is because L_B varieties are effectively surrounded by L_A varieties.[19] Therefore, only the contact situations in (33) and (35) play a role in our data, which explains why 3rd person subject clitics change more in the context of weather verbs than in referential contexts.[20] Now, consider 2nd singular subject clitics. In our dataset, their distribution is quite similar to the distribution of 3rd singular subject clitics in referential contexts: when one is present, the other is usually present too.[21] As much as for 3rd singular referential clitics, then, no overlap situation supporting change should be attested. This is in line with our data, which show that the rate of change of 2nd singular subject clitics and 3rd singular referential subject clitics is virtually identical.

As already seen, sentential negation is encoded by two features: preverbal negation (*neg preV*) and postverbal negation (*neg postV*), see (1)-(2). Both postverbal and preverbal negation are resistant to change. However, preverbal negation resists more than postverbal negation (and is the most resistant feature overall, see Plot 3). All three possible combinations of these two features are attested in Italo-Romance: only *neg preV* (central and southern varieties, plus Veneto, Friuli and Liguria), only *neg postV* (north-western Gallo-Italic varieties), both *neg preV* and *neg postV* (so-called "discontinuous" negation, mostly in between the previous two, in Emilia and in eastern Lombardy). The directionality of change follows the so-called Jespersen Cycle (Jespersen 1938, see Larrivée & Ingham 2011 for an overview): postverbal negation gains ground while preverbal negation retreats. In other words, we almost only see instances of varieties going from preverbal negation only (step I of the cycle) to discontinuous negation (step II), and varieties going from discontinuous negation (step II) to postverbal negation only (step III).[22] Since in our data preverbal negation is less prone to change than postverbal negation, step I → II happens more often than step II → III. To define potential overlaps, we must consider different contexts in which the same negation items are adopted. An obvious candidate is so-called *emphatic* negation (i.e. a negation expressing a high speaker commitment to $\neg p$, Breitbarth et al. 2013; see also Mattiuzzi & Poletto 2026 for a recent analysis of the relevance of emphatic negation markers for the diachrony of sentential negation). In Italian and in the Italo-Romance varieties where sentential negation is only marked by a preverbal clitic, emphatic negation is generally signalled by the addition of a postverbal item, see *mica* 'crumble' in (36).

(36) *Ma non ho mica incontrato nessuno!* (It.)
but not have.1SG mica met anyone
'Actually, I didn't meet anyone!'

[19] The same inspection was performed for possessives. In that case, all contact situations are consistently attested. The map can be found in the OSF folder (see fn. 5).

[20] Furthermore, consider that 3rd singular subject clitics in referential contexts are also supported by the presence of subject clitics for other persons (e.g. 2nd singular). In the contact situation in (34), the presence of such referential clitics might act as an anchor for the use of 3rd singular subject clitics in referential contexts. A full analysis of the patterns attested for all persons is therefore necessary to fully assess the role of overlap in this context. We leave this for future work.

[21] In some Venetan and Tuscan varieties, 2nd singular subject clitics extend more than 3rd singular subject clitics (cf. Renzi & Vanelli 1983). Their distribution is however more restricted than that of 3rd singular referential clitics in some Ladin and Provençal varieties (cf. the AIS datapoints Tuenno and Pontechianale). The distributional asymmetry between 2nd singular and 3rd singular referential subject clitics is however quantitatively negligible compared to the asymmetry between 3rd singular subject clitics in the context of weather verbs and in referential context.

[22] Finding apparent counterexamples to the cycle in this kind of data is not informative, as it is highly plausible that the range of data provided by the speakers incorporates a degree of variation among different grammars.

A sentence like (36) cannot be uttered out of the blue, as it manages previous assumptions by the other interlocutors (see Cinque 1976 on Italian *mica* and its distribution). Emphatic negation constitutes then an additional context that can act as an intra-linguistic support for CLI effects. To check that, we must first define how emphatic negation is generally marked in the Italo-Romance varieties. The most widespread pattern is the "Italian" one in (36), whereby regular sentential negation is marked by a preverbal clitic, while emphatic negation adds a postverbal item like *mica*. In varieties where regular negation already includes a postverbal item – whether preceded by a preverbal marker or not – two main possibilities arise: emphatic negation is either surface-identical to its regular counterpart or is signalled by a different postverbal item, alternative to the one that is used to convey regular sentential negation (see D'Antuono & Poletto 2026 for examples). Varieties where both regular and emphatic negation are preverbal are not attested, as well as varieties where emphatic negation is marked by the absence of a postverbal item which is otherwise involved in the expression of regular negation (the inverse of (36)). To the best of my knowledge, also varieties where regular negation is marked by a preverbal and a postverbal item, while emphatic negation is marked by the sole presence of the postverbal item are not present.[23] Considering this, we only have three contact situations (37)-(39).[24]

(37)

	L_A		L_B
regular	PREV	⟵	PREV+POSTV
emphatic	PREV+**POSTV**		PREV+POSTV

(38)

	L_A		L_C
regular	PREV	⟵	POSTV
emphatic	PREV+**POSTV**		POSTV

(39)

	L_B	L_C
regular	PREV+POSTV	POSTV
emphatic	PREV+POSTV	POSTV

Only (37) and (38) show a pattern of CLI supported by an intra-linguistic overlap. In both cases, it concerns the addition of a postverbal negative item to the regular negation context, from L_B/L_C to L_A. No pattern in favour of dropping the postverbal negative item is attested, as well as no pattern in favour of either dropping or adding the preverbal negative item. The only supported pattern of CLI is then the one favouring the passage from stage I to stage II of the Jespersen cycle, in line with our findings regarding the different resistance to change between *neg preV* and *neg postV*.
We now consider the remaining features classified as "overchangers" or "resistant".

[23] Rare patterns where emphatic negation is signalled by the addition of a preverbal clitic marker that is otherwise absent from the expression of regular negation are attested in Emilia (more precisely, in the Mantua area; see D'Antuono & Poletto 2026). Their rarity makes them however irrelevant for the present investigation.
[24] Varieties where regular and emphatic negation are surface-identical and varieties that adopt distinct postverbal negation markers for the two contexts are not distinguished in this context. This should not have a crucial impact for defining the overlap patterns, as in both cases the same overlap effects favouring the introduction of the postverbal item should be at play.

Besides postverbal wh-items, which are treated separately in Section 4.4, the remaining "overchangers" are auxiliary selection with reflexive clitic (*aux si*) and the use of Partitive Articles in indefinite contexts (*ind-obj-PA*).
Auxiliary selection – the alternation between different auxiliaries (typically versions of *have* and *be*) across different verb-classes – is well-known for being subject to complex implicational scales (see Sorace 2000). Furthermore, it can also be conditioned by other factors, such as the subject's person features (see Manzini & Savoia 2005:728, D'Alessandro & Roberts 2010, among many others). In practice, this means that most varieties necessarily present *extensive overlap patterns* in support of either direction of change (*have* → *be* or *be* → *have*), as they are both attested in many different contexts besides the one we consider here (verbs with a reflexive clitic). While a full analysis of such patterns cannot per performed, as it would require a full evaluation of the distribution of *have* and *be* auxiliaries in each variety, we can submit that *aux si* is subject to a high level of intra-linguistic overlap in support of both directions of change. This correctly predicts its high diachronic variability, comparable to the variability attested for *def-poss-pl-kin*.
In the Italo-Romance domain, indefinite Partitive Articles (PAs) compete with Bare Nouns, bare *de/di* 'of' preposition, definite articles, and other types of indefinite quantification, as pseudo-partitive constructions (see Cardinaletti & Giusti 2018; Pinzin & Poletto 2021a, 2022; Pinzin 2024). Their distribution is also conditioned by sentence polarity (Battye 1990; Garzonio & Poletto 2020). For the current analysis, a simplification has been introduced: the distribution of PAs has been made binary, where 1 corresponds to the presence of a PA, while 0 corresponds to the presence of *any of the other competitors*. This makes it impractical to compute partial overlap patterns, as they involve different alternatives for each variety. A more detailed analysis is therefore left for future research. However, it is worth mentioning that the use of indefinite PAs and Bare Nouns in Italo-Romance dialects has already been found to be subject to substantial across-languages priming effects in Italian-Dialect translation tasks (Pinzin & Poletto 2021b). If structural priming underlies CLI effects (see the discussion below), this might explain their high sensitivity to contact-induced change.
The remaining "resistant" feature is *dove-for-da*, exemplified in (40). (40a) employs the P *a* 'to' while (40b) the adverbial locative wh *addọ́* 'where'. (40b) can be analysed a free relative clause without an overt finite verb (corresponding to 'where the doctor (is)', see Munaro & Poletto 2013, 2014).

(40)
a. *a lu myẹ̄́riku* (Omignano, Cilentano, AIS)
to the doctor
b. *addọ́ lu myẹ̄́riku* (Acerno, Neapolitan, AIS)
where the doctor
'to the doctor'

Change from (40b) to (40a) is only attested for four datapoints in the area between Avellino and Benevento. Crucially, however, this is the only feature which is not attested in the MS database. This plays a role in explaining why its diachronic delta is so small. In fact, as soon as we only consider the ASIt data, its delta falls back in the range predicted by its breadth of contact.
We can conclude that within single groups, the number of overlaps conditions the relative rate of change. The realisation of the definite article in combination with possessives is declined in three contexts, depending on the N. Two contexts (singular kinship Ns, regular Ns) present a single overlap, one (plural kinship Ns) presents multiple overlaps. This is due to its central position in the implicational scale, see (24)-(29). Accordingly, the first two context change

less than the third. The same is valid for subject clitics, where 3rd singular clitics with weather verbs change more than referential 3rd and 2nd singular subject clitics, see (33)-(35). This is reflected in the presence of an overlap for the first feature, whereas the other two show no overlap (once actual geographical distribution is considered). Negation fits the pattern too: preverbal negation has no overlaps, contrary to postverbal negation, see (38)-(40). Indeed, preverbal negation changes less than postverbal negation. Other features, such as auxiliary selection, are amenable to similar analyses. However, while counting overlaps explains *within-group* asymmetries, *inter-group* comparison requires additional qualifications. Negation, the realisation of definite articles with possessives, and auxiliary selection behave as predicted: postverbal negation has one overlap and, coherently, changes comparably to *def-poss-kin*, *definite-possessive*, which have the same number of overlaps. Preverbal negation has no overlaps and changes comparatively less, while *def-poss-pl-kin* has multiple overlaps and changes more than all other features. Auxiliary selection fits too: it is subject to multiple overlaps and therefore changes as much as *def-poss-pl-kin*. However, subject clitics behave differently. Given equal numbers of overlaps, they change more than the other features. 3rd singular subject clitics with weather verbs should fall in the same class as *def-poss-kin*, *definite-possessive*, and postverbal negation (one overlap each). However, they change at similar rates as *def-poss-pl-kin* and auxiliary selection with *si*. 3rd and 2nd singular referential subject clitics should fall in the same class as preverbal negation, as all of them count zero patterns of overlap. Instead, they change as much as postverbal negation, *def-poss-kin*, and *definite-possessive*, which have one overlap. Whether this effect can be explained by other factors conditioning CLI, such as the interaction between subject realisation and information structure and/or frequency in the system, is left for future investigation.
From a more general perspective, this finding supports the idea that contact-induced change at the societal level is a function of contact within the individual. This can be conceptualised in the following terms. The locus where languages are in contact are multilingual speakers, who store in their linguistic capacity alternative grammatical rules, that is, rules that occupy the same grammatical space. A bilingual speaker of Italian and English, for example, stores both the English rule that produces A-N (*red rose*) and the complementary N-A rule that is relevant for Italian in the same context (*rosa rossa*, lit. 'rose red'). The activation of either of the two grammatical rules, whether in processing or production, depends on the context of utterance: an Italian context activates N-A, an English context activates A-N. Evidently, the same is valid for bidialectal speakers and each of the 23 rules considered here. Contexts are permeable, however, as abundant records of CLI effects show. A rule from L_A can be activated within a context that is otherwise mainly characterisable as L_B, and *vice versa*. Over time, this can lead to longstanding changes in the context of activation of either of the two alternative grammatical rules within the individual. Generalisation to the societal level can then result in language change (see Muysken 2010 for an in-depth characterisation of the scenarios for contact). By showing how restrictions applying to CLI also modulate the effects of contact-induced change, this study directly links individual-level processes with diachronic contact-induced change. However, CLI effects are themselves only secondary outcomes of other cognitive processes, such as structural priming (see Serratrice 2016), which are then the crucial modulators of contact-induced change. Indeed, structural priming has already been connected to contact-induced change, both theoretically (Jäger & Rosenbach 2008, Pickering & Garrod 2017) and empirically (Kootstra & Sahin 2018, Jacob et al. 2025, Karkaletsou et al. 2025, Baroncini et al. 2026, and the collection in Engemann & Şafak 2026). From this perspective, our data lay the ground for testing the full extension of such connection, from contact-induced change to structural priming, through CLI effects.

4.4 Innovations without contact

In the Italian dialectal context, contact is a highly significant predictor of change, and structural compatibility explains much of the remining variation. Nevertheless, contact-independent variation cannot be ruled-out. The reason is not only conceptual (to which I return below) but also empirical: postverbal wh-items – exemplified in (41) – only developed at t_1, without being ever attested at t_0. To the extent that this holds, this is evidence to maintain that change can also happen without contact as a trigger.

(41)

a. *Cos* *ti* *è* *fec?* (Brione, Lombardy, ASIt)
what SBJ.CL.2SG have done

b. *G'* *het* *fat* *chi?* (Iseo3, Lombardy, ASIt)
LOC.CL have.2SG done what
'What did you do?'

Plausible explanations might appeal to spontaneous innovations or language-internal triggers. In this respect, it is worth noting that the notion itself of language-internal trigger is theoretically challenging. Speakers' competence (or i-language) reflects universal constraints on how grammars are built (First factor), external input (Second factor), and pressure from general cognitive constraints (Third factor), see Chomsky (2005), among many others. Since First and Third factors are invariant, variation in the input is the classical *locus* where changes in the competence can arise. The view of language-contact adopted here, defined as the presence in the speaker's competence of rules that occupy the same grammatical space, is fully compatible with this idea: using one or the other rule depends on their contextual degree of activation, which is in turn a function of the input. Contact-induced variation in the input – a language-external issue – can then determine changes in the rules' distribution. Likewise, spontaneous innovations are fully compatible with the general framework. They appeal to a certain degree of freedom that speakers have in defining the distribution of the items of the language. This degree of freedom can translate into novel grammatical rules, which then enter in competition with what already occupies that same grammatical space (who triggers and spreads the innovation, and where, is immaterial to the issue).[25] The range within which such innovations can appear is necessarily constrained by the structure of the grammar (see the implicational scales above, for example). The crucial point, however, is that while such innovations are grammatically constrained, there is no language-internal mechanism *triggering* them nor, therefore, any necessary path of change. All in all, with both contact and innovation change relates to the contextual use of a given grammatical rule due to competition with another rule, either an already existing one (contact) or a new one (innovation). Competition could then be modelled via learning algorithms (see Yang 2000 and ff.; see also Truswell and Lassiter 2025 for a recent re-evaluation and an updated model). Language-internal processes of change – i.e. processes of change that are triggered by the internal mechanics of the system – are instead not easily modellable under the standard view of competence sketched above. Being internal, they necessarily relate to either the First or the Third factor. This however runs into the risk of defining teleological paths of change, if no balance among different language-internal pull factors is posited (but see Van Gelderen 2021 for a recent proposal and a review of the previous literature on possible changes due to efficiency of computation, a Third factor process). Be it as it may, the relevant point in this context is that the source of postverbal wh-items is not due to contact and is either to be

[25] This is irrelated to cases of "mislearning" during acquisition, which are extremely dubious as triggers for language change (see Yang 2000b: 237, but also Meisel et al. 2013, among many others).

connected to spontaneous innovation or to a process of change triggered by language-internal factors. Nothing in our data pushes for either option.

4.5 Areal differences

Before concluding, we investigate possible differentials across dialectal areas. It is indeed plausible that contact-induced change proceeds faster in some areas than in others, due to specific conditions (for a factorisation of such conditions in terms of similarity, prestige, proficiency, number of speakers, time, attitude, see Muysken 2013: 726). This would push change for the features whose borders lie in those areas, compared to the other features. In other words, we need to put to test what was said in Section 1, where dialectal environments were defined as predominantly uniform. To control for this, we checked *via* a post-hoc analysis whether features for which contact is mostly attested in a given area A tend to change more than the others. We leveraged the subdivision in linguistic areas proposed in Pellegrini (1977) and counted for each feature F how many contact-linked datapoints (= datapoints that are close to each other and have a different value for F, see (5)) are present in each of the 13 dialectal Macro-Areas. This derives the share of contact-linked datapoints per Macro-Area for F ($C\%_{F,A}$ = contact share for F in A). To check if a higher share in A a is a relevant factor for predicting the feature's attested change (Δ_F), we consider each Macro-Area at a time.[26] For each Macro-Area A, we could in principle directly regress Δ_F on the share of contact-linked datapoints for F in A ($C\%_{F,A}$). This would tell us how much of the variation in Δ_F can be explained by a high concentration of F's contact zone in A. However, a high $C\%_{F,A}$might merely reflect the overall breadth of contact attested for F (B_F). To control for this, we need to fit a regression with both $C\%_{F,A}$ and B_F as predictors. Out of the 13 Macro-Areas, only Veneto adds an almost significant factor to the simple regression that only considers B_F (Table 5).[27]

feature	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	-0.0146877	0.0184727	-0.795	0.436
B_F	0.0014150	0.0003421	4.136	<.001 ***
$C\%_{F,A}$ (A = Veneto)	0.3656161	0.1966697	1.859	0.079 .

Table 5. Regression results, predictor B_F and $C\%_{F,A}$ (Veneto)

This indicates that features with a high contact share in Veneto tend to change faster than features with a low contact share in this Macro-Area, even if the effect is not particularly strong. Furthermore, with only 23 phenomena, we cannot exclude that the effect is driven by a small number of features that both concentrate their contact in Veneto and change rapidly for independent reasons. A larger dataset would be needed to establish whether the area itself accelerates change independently of the features it hosts. The analysis has been replicated adopting two further grouping systems: administrative regions (22 regions) and regional groups.[28] In both cases, no single region/regional group stands out. Only Friuli Venezia Giulia and Veneto reach partial significance, as well as the North-East group. This is because they include the signal from the Veneto Macro-Area (some parts of Friuli Venezia-Giulia belong to

---

[26] We test each Macro-Area separately because the shares of the 13 area sum to 1, making the joint regression not identified.

[27] R^2: 0.720 (Adjusted R-squared: 0.6908). Analysis of Variance (ANOVA) shows how the new model is not significantly better (Pr(>F): 0.07858) than the simple model with just B_F as a predictor.

[28] North-East = Trentino Alto Adige + Veneto + Friuli Venezia Giulia; North-West = Lombardia + Piemonte + Grisons + Valle d'Aosta + Liguria + Emilia; Central = Toscana + Marche + Lazio + Umbria + Molise + Abruzzo; Southern = Campania + Basilicata + Puglia + Calabria + Sicilia; Sardinia.

the Veneto Macro-Area). All in all, the areal analysis shows that only Veneto (partially) stands out, while all other areas do not have any significant effect on Δ_F. This suggests that contact-induced change is predominantly homogeneous across Italy, highlighting the relative uniformity of the environment.

5. Conclusions and future work

This study proposed a novel methodology to quantify the role of contact in explaining patterns of language change. The results are striking: the simple count of the neighbouring datapoints that express a different value for a given feature at t_0 explains almost 66% of the feature's attested diachronic variation between t_0 and t_1. In other words, how much a given feature changes through time is, to a great extent, a function of the amount of contact it is exposed to. However, the analysis also shows that not all variation can be explained as a simple function of contact. In this respect, a post-hoc qualitative analysis showed that resistance to contact-induced change prevalently depends on the degree of structural compatibility between the languages in contact. An operationalisation of structural compatibility in terms of number of partial overlaps supporting CLI effects (Hulk & Müller 2000) indeed shows that structures with intra-linguistic support are transferred more easily. On the contrary, general borrowability scales do not appear to play any role. Overall, our results show that language change is highly dependent on language contact and that, among plausible language-internal grammatical factors, only structural compatibility is crucial, in terms of CLI effects. This is in line with the idea that contact-induced change is a function of individual-level processes active within multilingual speakers.

Some limitations of the study deserve mentioning. An obvious one regards generalizability beyond the specific environment considered. Our conclusions are valid for the Italian dialectal environment and the features investigated. To improve generalizability, further areas and features must be integrated (e.g. in the direction of the German-speaking area, see Weiß & Ströbel 2024). Indeed, the role of contact might well be amplified within our dialectal set-up, where neighbouring varieties are generally grammatically similar to each other and not subject to overt schooling. This might boost the role of contact as a driver of change, while at the same time downsizing the role of other contact-independent factors. A second potential limitation relates to the time-span. Change is here calculated in the space of around 100 years, a relatively short amount of time. Potential contact-independent factors might need more time to be visible, leading to an underestimation of their role.

The results of the study naturally lead to further developments. A yet unexplored line of inquiry concerns change directionality. The number of partial overlaps supporting CLI effects in the contact situations attested for each feature define an expected direction of change. To exemplify, negation is only expected to change in the direction of the expansion of postverbal negation, while for preverbal negation no intra-linguistic pull factors are detected. This is because, in the attested contact situations, one pattern of overlap favours the expansion of postverbal negation, while no pattern of overlap concerns preverbal negation. Though this is confirmed by the data we have, these are rather limited and further analyses are necessary, also considering other features. As a second line of inquiry, suggested in Section 4.2, these data lend themselves to further testing the connection between language change, CLI effects, and structural priming. Our data already demonstrate a connection between CLI and contact-induced change. A natural extension is then to adopt a priming paradigm to check if the same relative strength of change/CLI can be reproduced experimentally in the same structural conditions. Furthermore, if directionality is confirmed to follow from the definition of overlap patterns in the contact situations, it becomes possible to experimentally reproduce such expected directional changes within a priming experiment (as already suggested in Jäger & Rosenbach 2008).

Acknowledgements
[To Complete]

Funding
[To Complete]

References

Aikhenvald, Alexandra Y. 2006. Grammars in Contact A Cross-Linguistic Perspective. In Alexandra Y. Aikhenvald & R. M. W. Dixon (eds), *Grammars in Contact: A Cross-Linguistic Typology*, 1-66. Oxford: Oxford University Press https://doi.org/10.1093/oso/9780199207831.003.0001

Baroncini, Ioli, Anna Michelotti & Helen Engemann. 2026. Priming motion events in Italian heritage language speakers: Agents and mechanisms of language change. *Linguistic Approaches to Bilingualism* 16(4), 429-456.

Battye, Adrien. 1990. La quantificazione nominale, il veneto e l'italiano a confronto con il genovese e il francese. *Annali di Ca' Foscari* 29, 27–44.

Benincà, Paola & Cecilia Poletto. 2007. The ASIS Enterprise: A View on the Construction of a Syntactic Atlas for the Northern Italian Dialects. *Nordlyd*. 34(1).

Bidese, Ermenegildo & Alessandra Tomaselli. 2021. Language synchronization north and south of the Brenner Pass: modeling the continuum. STUF – *Language Typology Universals* 74: 185-216. https://dx.doi.org/10.1515/stuf-2021-1028

Breitbarth, Anne, Karen De Clercq & Liliane Haegeman 2013. The syntax of polarity emphasis. *Lingua* 128. 1–8.

Cardinaletti, Anna & Giuliana Giusti. 2016. The syntax of the Italian indefinite determiner dei. *Lingua* 181. 58-80. https://doi.org/10.1016/j.lingua.2016.05.001

Cardinaletti, Anna & Giuliana Giusti. 2018. Indefinite determiners, variation and optionality in Italo-Romance. In Roberta D'Alessandro and Diego Pescarini (eds.), *Advances in Italian Dialectology*, 135-161. Leiden/Boston: Brill. https://doi.org/10.1163/9789004354395_008

Cardinaletti, Anna and Giusti, Giuliana. 2020. Indefinite determiners in informal Italian: A preliminary analysis. *Linguistics* 58 (3), 679-712. https://doi.org/10.1515/ling-2020-0081

Chomsky, Noam. 2005. Three factors in language design. *Linguistic Inquiry* 36(1), 1–22.

Cinque, Guglielmo. 1976. mica. *Annali della Facoltà di Lettere e Filosofia dell'Università di Padova* 1, 101–112.

Craenenbroeck, Jeroen Van, & Koppen, Marjo Van. 2025. Quantitative Approaches to Syntactic Variation. In Sjef Barbiers, Norbert Corver & Maria Polinsky (eds.), *The Cambridge Handbook of Comparative Syntax*, 108-140. Cambridge: Cambridge University Press.

D'Alessandro, Roberta & Ian Roberts 2010 Past participle agreement in Abruzzese: split auxiliary selection and the null-subject parameter. *Natural Language and Linguistic Theory* 28, 41–72. https://doi.org/10.1007/s11049-009-9085-1

D’Antuono, Nicola & Cecilia Poletto. 2026. The pragmatic functions of negative concord and doubling in Mantuan. Paper presented at *Incontro di Grammatica Generativa 51* (IGG51, 12th–14th February 2026, University of Udine).
Engemann, Helen & Duygu F. Şafak. 2026. Psycho-historical linguistics: a new empirical agenda linking bilingual processing, acquisition, and change. *Linguistic Approaches to Bilingualism* 16(4), 397-401. https://doi.org/10.1075/lab.26046.eng
Gardani, Francesco. 2022. Contact and Borrowing, in Adam Ledgeway and Martin Maiden (eds.) *The Cambridge Handbook of Romance Linguistics*, 845–869. Cambridge: Cambridge University Press.
Garzonio, Jacopo & Cecilia Poletto. 2020. Partitive objects in negative contexts in Northern Italian dialects. *Linguistics* 58. 621-650. https://doi.org/10.1515/ling-2020-0079
Gelderen Elly Van. 2021. *Third factors in language variation and change*. Cambridge: Cambridge University Press.
Hulk, Aafke & Natasha Müller. 2000. Bilingual first language acquisition at the interface between syntax and pragmatics. *Bilingualism: Language and Cognition* 3, 227–244.
Jaberg, Karl & Jakob Jud. 1928-1940. *Sprach- und Sachatlas Italiens und der Südschweiz. Vol. 1-8*. Zofingen: Ringier.
Jacob, Gunnar, Hanife Ilen & Helen Engemann. 2025. The role of cross-linguistic structural priming in contact-induced language change: Ungrammatical comparative priming in Turkish–German bilinguals. *Linguistic Approaches to Bilingualism*. https://doi.org/10.1075/lab.24055.jac
Jakobson, Roman. 1938. Sur la théorie des affinités phonologiques entre les langues. In K. Barr, Viggo Brøndal, Louis L. Hammerich & Louis Hjelmslev. (eds), *Actes du Quatrième Congrès International de Linguistes, tenu à Copenhague du 27 août au 1er septembre 1936*. Copenhagen: Munskgaard, 48–59.
Jäger, Gerhard & Anette Rosenbach. 2008. Priming and unidirectional language change. *Theoretical Linguistics* 34 (2): 85-113.
Jeszenszky, Péter, Philipp Stoeckle, Elvira Glaser & Robert Weibel. 2017. Exploring Global and Local Patterns in the Correlation of Geographic Distances and Morphosyntactic Variation in Swiss German. *Journal of Linguistic Geography* 5(2), 86–108. https://doi.org/10.1017/jlg.2017.5.
Henri Kauhanen, Deepthi Gopal, Tobias Galla & Ricardo Bermúdez-Otero. 2021. Geospatial distributions reflect temperatures of linguistic features. *Science Advances* 7eabe6540. https://doi.org/10.1126/sciadv.abe6540
Karkaletsou, Foteini, Gunnar Jacob & Shanley E. M. Allen. 2025. Cross-linguistic structural priming of innovations in Canadian French: Evidence from a language contact situation. *Linguistic Approaches to Bilingualism* (online first). https://doi.org/10.1075/lab.24100.kar
Kootstra, Gerrit J. & Hülya Şahin. 2018. Crosslinguistic structural priming as a mechanism of contact-induced language change: Evidence from Papiamento–Dutch bilinguals in Aruba and the Netherlands. *Language* 94(4), 902–930.
Kouteva, Tania, Bernd Heine, Bo Hong, Haiping Long, Heiko Narrog, and Seongha Rhee. 2020. *World Lexicon of Grammaticalization. 2nd ed*. Cambridge: Cambridge University Press.
Larrivée Pierre 2011. Is there a Jespersen cycle? In Larrivée Pierre & Richard Ingham (eds.), *The Evolution of Negation: Beyond the Jespersen Cycle*, 1-22. Berlin: de Gruyter Mouton.
Loporcaro, Michele, Stephan Schmid, Chiara Zanini, Diego Pescarini, Giulia Donzelli, Stefano Negrinelli & Graziano Tisato. 2021. AIS, reloaded: A digital dialect atlas of Italy and southern Switzerland. In: Thibaut, André, Matthieu Avanzi, Nicolas Lo

Vecchio and Alice Millour (eds). *Nouveaux regards sur la variation dialectale*, 111–136. Strasbourg: Editions de Linguistique et de Philologie.
Madaro, Romano, Cosentino, Michele, Bidese, Ermenegildo & Simone Barco. 2025. Clitic syntax in areal linguistics: Pinning down the role of language contact via crowdsourcing. *Isogloss* 11(6)/6, 1-24. https://doi.org/10.5565/rev/isogloss.514
Manzini & Savoia 2005a. *I dialetti italiani e romanci. Morfosintassi generativa. Vol 1*. Firenze: Edizioni dell'Orso.
Manzini & Savoia 2005b. *I dialetti italiani e romanci. Morfosintassi generativa. Vol 2*. Firenze: Edizioni dell'Orso.
Manzini & Savoia 2005c. *I dialetti italiani e romanci. Morfosintassi generativa. Vol 3*. Firenze: Edizioni dell'Orso.
Mattiuzzi, Tommaso & Cecilia Poletto. 2026. There Is no First Phase of the Jespersen Cycle. *Transactions of the Philological Society*. https://doi.org/10.1111/1467-968x.70033
Meisel, Jürgen, Martin Elsig & Esther Rinke. 2013. *Language Acquisition and Change: A Morphosyntactic Perspective*. Edinburgh: Edinburgh University Press.
Mensching, Guido & Eva-Maria Remberger. 2016. Sardinian. In Adam Ledgeway & Martin Maiden, (eds.), *The Oxford guide to the Romance languages*, 270-318. Oxford: Oxford University press.
Müller, Natasha & Aafke Hulk. 2001. Crosslingusitic influence in bilingual language acquisition: Italian and French as recipient languages. *Bilingualism: Language and Cognition*, 4, 1–21.
Munaro, Nicola & Cecilia Poletto. 2013. Indizi interlinguistici sulla struttura interna dell'elemento interrogativo 'dove' nei dialetti italiani settentrionali, *Quaderni di lavoro ASIt* 15, 1-20.
Munaro, Nicola & Cecilia Poletto. 2014. Synchronic and diachronic clues on the internal structure of 'where' in Italo-Romance. In Paola Benincà et al. (eds.), *Festschrift in honor of M Parry*, 279-300. Oxford & New York: Oxford University Press.
Muysken, Peter. 2010. Scenarios for Language Contact. In Raymond Hickey (ed.), *The Handbook of Language Contact*, 265–281. Chichester: Wiley-Blackwell.
Nerbonne, John. 2010. Measuring the diffusion of linguistic change. *Philosophical Transactions of the Royal Society of London*. Series B, Biological Sciences 365(1559). 3821–8. doi:10.1098/rstb.2010.0048.
Nerbonne, John. 2013. How much does geography influence language variation? In Peter Auer, Martin Hilpert, Anja Stukenbrock and Benedikt Szmrecsanyi (eds.), *Space in Language and Linguistics: Geographical, Interactional, and Cognitive Perspectives*, 222-239. Berlin, Boston: De Gruyter. https://doi.org/10.1515/9783110312027.222
Nerbonne, John & Wilbert Heeringa. 2007. Geographic Distributions of Linguistic Variation Reflect Dynamics of Differentiation. In Sam Featherston & Wolfgang Sternefeld (eds.), *Roots: Linguistics in Search of its Evidential Base*, 267–298. Berlin: De Gruyter Mouton.
Pellegrini, Giovan Battista. 1977. *Carta dei dialetti d'Italia*. Pisa: Pacini.
Pickering, Martin J. & Simon Garrod. 2017. Priming and Language Change. In Marianne Hundt, Sandra Mollin & Simone E. Pfenninge (eds.), *The Changing English Language: Psycholinguistic Perspectives*, 173–190. Cambridge: Cambridge University Press.
Pinzin, Francesco. 2024. What's hidden below definiteness and genitive: on indefinite partitive articles in Romance. *Linguistics* 62(5): 1251-1300. https://doi.org/10.1515/ling-2022-0059
Pinzin, Francesco & Cecilia Poletto. 2021a. Articoli partitivi e sintagmi nominali indefiniti: distribuzione comparativa nelle lingue dell'Italia del nord. *Rivista Italiana di Dialettologia* 45(1), 1-54.

Pinzin, Francesco & Cecilia Poletto. 2021b. Indefinite objects in micro-variation. A cross-linguistic analysis of the distribution of partitive articles, bare nominals and definite determiners in Northern Italy. *Studia Linguistica* 76(1), 13-55. http://doi.org/10.1111/stul.12187
Pinzin, Francesco & Cecilia Poletto. 2022. An indefinite maze: on the distribution of partitives and bare nouns in the Northern Italian dialects. *Isogloss* 8(2)/20, 1-23. https://doi.org/10.5565/rev/isogloss.130
Poletto, Cecilia. 2000. *The higher functional field: Evidence from Northern Italian Dialects*. Oxford: Oxford University Press.
R Core Team. 2025. *R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing*, Vienna, Austria. URL https://www.R-project.org/.
Renzi, Lorenzo & Laura Vanelli. 1983. I pronomi soggetto in alcune varieta romanze. In *Scritti linguistici in onore di Giovan Battista Pellegrini*, 121-145. Pisa: Pacini.
Roberts, Ian & Anna Roussou. 2003. *Syntactic Change: A Minimalist Approach to Grammaticalization*. Cambridge: Cambridge University Press.
Rohlfs, Gerhard. 1969. *Grammatica storica della lingua italiana e dei suoi dialetti: Sintassi e formazione delle parole*. Torino: Einaudi.
Savoia, Leonardo M., Maria R. Manzini, Greta Mazzaggio, Ludovico Luca Andrea, Mael V. Vena, Carlo Zoli, Benedetta Baldi, Ludovico Franco, Neri Binazzi & Achille Fusco. (2026). *The Manzini & Savoia (2005) Corpus: Morphosyntactic Variation in Italian and Romansh Dialects (2.0)* [Data set]. Zenodo. https://doi.org/10.5281/zenodo.19352290
Séguy, Jean. 1971. La relation entre la distance spatiale et la distance lexicale. *Revue de Linguistique Romane* 35(138), 335–357.
Serratrice, Ludovica. 2013. Cross-linguistic influence in bilingual development: Determinants and mechanisms. *Linguistic Approaches to Bilingualism*, 3(1), 3–25. https://doi.org/10.1075/lab.3.1.01ser
Serratrice 2016. Cross-linguistic influence, cross-linguistic priming and the nature of shared syntactic structures. *Linguistic Approaches to Bilingualism* 6(6), 822–827.
Sorace, Antonella. 2000. Gradients in Auxiliary Selection with Intransitive Verbs. *Language* 76(4), 859–90. https://doi.org/10.2307/417202.
Sorace, Antonella & Francesca Filiaci. Anaphora Resolution in Near-Native Speakers of Italian. *Second Language Research* 22(3), 339–68. http://www.jstor.org/stable/43103710.
Szmrecsanyi, Benedikt. 2012. Geography is overrated. In Sandra Hansen, Christian Schwarz, Philipp Stoeckle & Tobias Streck (eds.), *Dialectological and Folk Dialectological Concepts of Space - Current Methods and Perspectives in Sociolinguistic Research on Dialect Change*, 215–231. Berlin, Boston: De Gruyter.
Thomason, Sarah Grey & Terrence Kaufman. 1988. *Language Contact, Creolization, and Genetic Linguistics*. 1st ed. Oakland: University of California Press. https://doi.org/10.2307/jj.8501341.
Tomaselli, Alessandra & Ermenegildo Bidese. 2023. Fortune and Decay of Lexical Expletives in Germanic and Romance along the Adige River. *Languages* 8: 44. https://doi.org/10.3390/languages8010044
Truswell, Robert & Daniel Lassiter. 2025. Failed changes and the Superset Problem: The importance of learning in models of syntactic change. *Lingbuzz*, lingbuzz/009556.
Unsworth, Sharon. 2003. Testing Hulk & Müller (2000) on crosslinguistic influence: Root Infinitives in a bilingual German/English child. *Bilingualism: Language and Cognition* 6(2), 143–158. doi:10.1017/S1366728903001093.

Vergeiner, Philip C., Lars Bülow, and Stephan Elspaß. 2025. Analyzing Dialect (Morpho)Syntax in Austria: A Non-Aggregative Dialectometric Approach. *Journal of Linguistic Geography* 12(2), 57–68. https://doi.org/10.1017/jlg.2024.15.
Weinreich, Uriel. 1953. *Languages in contact: findings and problems*. The Hague: Mouton.
Weiß (2019): Rebracketing (Gliederungsverschiebung) and the Early Merge Prinicple. *Diachronica* 36(4), 509-545.
Weiß (2021): Reanalysis involving Rebracketing and Relabeling – a special type. *Journal of Historical Syntax*, 5(39), 1-26.
Weiß, Helmut & Thoms Ströbel. 2024. Neuere Entwicklungen in der Dialektsyntax. *Linguistische Berichte* 35, 47-80.
Wieling, Martijn & John Nerbonne. 2015. Advances in Dialectometry. *Annual Review Linguistics* 1, 243-264. https://doi.org/10.1146/annurev-linguist-030514-124930
Yang, Charles D. 2000a. *Knowledge and Learning in Natural Language*. Doctoral dissertation, Cambridge (Mass.): Massachusetts Institute of Technology.
Yang, Charles D. 2000b. Internal and external forces in language change, *Language Variation and Change*,12(3), 231–250. https://doi.org/10.1017/S0954394500123014.
Yip, Virginia & Stephen Matthews. 2000. Syntactic Transfer in a Cantonese–English Bilingual Child. *Bilingualism: Language and Cognition* 3(3), 193–208. https://doi.org/10.1017/S136672890000033X.